

LUMINA JEWELS — COMPREHENSIVE PROJECT REPORT & PRESENTATION GUIDE

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Project Name: Lumina Jewels Management System

Framework Stack: React, Vite, CSS, Firebase (Auth, Firestore, Storage)

Deployment Pipeline: Vercel Continuous Integration

Target Audience: Stakeholders, Technical Evaluators, and Presentation Panels

1. Project Overview & Business Objectives

Lumina Jewels is a next-generation, high-performance web platform designed to streamline operations for luxury jewellery retail brands. In the traditional jewellery retail sector, businesses face challenges managing catalog updates, adjusting to volatile daily precious metal rates (Gold, Silver, Platinum), coordinating bookings across multiple physical showrooms, and maintaining strict security standards.

Lumina Jewels solves these issues by offering a cohesive digital solution:

- Dynamic Valuation:** Integrates live commodity prices to recalculate product values instantly based on weight and purity.
- Branded Multi-Store Support:** Directs guests and employees to location-specific inventories and showroom booking systems.
- Low-Code CMS Integration:** Enables non-technical managers to update home page content, store details, and navigation bars.
- Enterprise Security Auditing:** Tracks system accesses, blocks fraudulent login attempts, and keeps records of administrative changes.

2. System Architecture & Tech Stack

The system is built on a highly modular, decoupled architecture optimized for speed, scalability, and security.

```
graph TD
    Client[React Frontend / Vite] -->|Auth & Queries| FirebaseSDK[Firebase client SDK]
    Client -->|Local API calls| LocalProxy[Vite Proxy Middleware]
    LocalProxy -->|Rates Request| ExternalAPI[GoldAPI.io API]
    FirebaseSDK -->|OAuth / JWT| Auth[Firebase Auth]
    FirebaseSDK -->|Data Sync| DB[(Cloud Firestore)]
    FirebaseSDK -->|Static Assets| Storage[(Cloud Storage)]
    Client -->|Hosting / SSL| Vercel[Vercel Edge Network]
```

Technology Breakdown

- Core Library:** React 19 (for high-speed UI rendering and robust hook state).
- Build System:** Vite (for fast build times, environment variable loading, and hot module reloading).
- Styling Engine:** Custom Vanilla CSS (configured with a modern dark-gold design token system, eliminating heavy dependencies like Tailwind).
- Backend Services:** Google Firebase (Serverless Architecture):
 - Firebase Auth:** Handles user tokens, validation, and session states.
 - Cloud Firestore:** Real-time NoSQL document store with custom security rules.
 - Cloud Storage:** Enforces asset upload limits and file extensions for images.
- Integrations:** GoldAPI.io (REST endpoint integrated via a Vite proxy to fetch live precious metal commodity rates).

3. Database Schema Design (Firestore NoSQL)

Below are the schema designs of the core Firestore collections:

3.1. users Collection

- Document ID:** Firebase User UID (auth.uid)

```
{
  "name": "Pragati Tayade",
  "email": "pragatitayade1302@gmail.com",
  "phone": "+91 98765 43210",
  "role": "superadmin",
  "addresses": [
    {
      "id": "1718520000000",
      "title": "Home Address",
      "street": "123 luxury Row, Showroom Blvd",
      "city": "Mumbai",
      "state": "Maharashtra",
      "pincode": "400001",
      "isDefault": true
    }
  ],
  "failedLoginAttempts": 0,
  "isBlocked": false,
  "status": "online",
  "joinDate": "Jun 2026",
  "createdAt": "2026-06-16T05:00:00.000Z",
  "lastCheckIn": "Timestamp(1718520000)"
}
```

3.2. stores Collection

- **Document ID:** Auto-generated ID (store_01)

```
{
  "name": "Khandelval Showroom",
  "code": "KHND",
  "address": "45 Jewellers Lane, Commercial Zone",
  "city": "Nagpur",
  "phone": "+91 712 345678",
  "status": "active",
  "createdAt": "2026-06-16T00:00:00.000Z"
}
```

3.3. loginActivity Collection

- **Document ID:** Auto-generated ID

```
{
  "userId": "auth_uid_here",
  "userName": "Pragati Tayade",
  "email": "pragatitayade1302@gmail.com",
  "role": "superadmin",
  "loginTime": "2026-06-16T05:42:00.000Z",
  "ipAddress": "103.88.22.41",
  "deviceInfo": "Mozilla/5.0 (Windows NT 10.0; Win64; x64)...",
  "status": "success",
  "errorMessage": ""
}
```

3.4. auditLogs Collection

- **Document ID:** Auto-generated ID

```
{
  "action": "USER_LOGIN",
  "module": "Auth",
  "actorId": "auth_uid_here",
  "targetId": null,
  "details": {
    "email": "pragatitayade1302@gmail.com",
    "role": "superadmin"
  },
  "timestamp": "2026-06-16T05:42:00.000Z"
}
```

4. Key Functional Modules

4.1. Core Authentication & Gatekeeping

Handles user management, registration, and log-ins across **seven** roles:

1. **superadmin** — Unlimited administrative control.
2. **admin** — Manages store locations, inventory, and staff rosters.
3. **manager** — Supervises daily store catalog and appointments.
4. **staff** — Manages stock and updates order tracking.
5. **finance** — Views revenue analytics and handles payments.
6. **delivery** — Manages and dispatches delivery orders.
7. **customer** — Accesses wishlists, custom orders, and addresses.

4.2. Multi-Store Selection Routing

On initial startup, guests can browse the website globally. When selecting products or booking custom appointments, the customer is prompted to select a physical store location (e.g., Khandelval). Staff members log directly into their specific branch dashboard.

4.3. Dynamic Pricing & Metal Rates Engine

Calculates the price of precious metals in real-time. Products are assigned a metal type (Gold, Silver, Platinum), a karat level (18K, 22K, 24K), weight in grams, and making charges.

$$\text{Price} = (\text{Rate per Gram} \times \text{Weight}) + \text{Making Charges} + \text{Tax}$$

If the GoldAPI service fails or runs out of credits, the engine switches to a local mock rate fallback to keep the catalog functional.

4.4. Low-Code Branding & CMS Dashboard

Administrators can edit the website's appearance through the Admin panel. This includes uploading logo files, changing the hero carousel, adding social media contact details, and modifying navigation links (like adding custom collections).

4.5. Cart & Order Fulfillment

Provides a seamless checkout process. The cart tracks items in a state synced to `localStorage`. Checkout automatically reserves the physical stock, calculates the correct GST, and records the order in Firestore.

5. Security Architecture

5.1. Firestore Rules Protection

Lumina Jewels uses strict Firebase security rules (`firestore.rules`) to protect against unauthorized database edits:

```
rules_version = '2';
service cloud.firestore {
  match /databases/{database}/documents {
    // Audit logs are read-only and write-only, deletion is blocked
    match /auditLogs/{log} {
      allow create: if request.auth != null;
      allow read: if request.auth != null && request.auth.token.role in ['superadmin', 'admin'];
      allow delete, update: if false;
    }
    // Users can only view or write their own documents
    match /users/{userId} {
      allow read, write: if request.auth != null && (request.auth.uid == userId || request.auth.token.role == 'superadmin');
    }
  }
}
```

5.2. Account Lockout Security

The system tracks invalid authentication requests in the `users` collection. If an email address registers 5 consecutive failed login attempts, the `isBlocked` boolean is toggled to `true`. This locks the account to prevent brute-force attacks until an administrator unlocks it.

6. Performance & Spacing Optimizations

6.1. Non-Blocking Logger Execution

To ensure fast page transition times, login auditing has been decoupled from the primary login flow:

- **Old Behavior:** The login process waited (`awaited`) for both the external IP tracking fetch and the Firestore audit log write to finish. A slow network connection could freeze the UI for 30+ seconds.
- **New Behavior:** The login script triggers the audit tracking function asynchronously without the `await` keyword. The user is redirected to their dashboard immediately (185ms), while logging executes in the background.
- **Timeout Shield:** An `AbortController` aborts the third-party IP fetch request after 1.5s if it hangs, preventing network issues from slowing down the app.

6.2. Responsive Spacing System

To handle custom navigation items (like "rings") cleanly, we optimized the main header CSS:

- **Intermediate Spacing:** Screens under 1400px compress nav link padding, shrink the search box, and reduce gaps to save horizontal space.
- **Adaptive Breakpoint:** Raising the mobile menu collapse breakpoint to 1440px ensures that the header layout cleanly switches to mobile format before elements overlap.
- **Preservation:** `flex-shrink: 0` is applied to `.header-actions` to prevent the search bar, cart, wishlist, and login buttons from shrinking or sliding off-screen.

7. Presentation Pitch Outline (Slide-by-Slide)

Use this outline for presentation slide preparation:

Slide 1: Cover & Branding

- **Title:** Lumina Jewels Management System
- **Subtitle:** Modern E-Commerce and Enterprise ERP for High-End Jewellery Retail
- **Presenter:** [Your Name]

Slide 2: The Business Problem

- Volatile commodity markets: Gold, silver, and diamond prices shift daily.
- Multi-channel friction: Managing inventory across multiple showrooms is difficult.
- Slow administrative adaptation: Changes to branding or navigation require manual coding.

Slide 3: The Solution

- Real-time metal rate calculation engine.
- Multi-store location routing.
- Low-code CMS control panel for administrators.
- Strict audit trails and secure login safeguards.

Slide 4: Tech Stack Overview

- **Frontend:** React, Vite, Custom CSS.
- **Backend:** Serverless Google Firebase (Auth, Firestore, Cloud Storage).
- **Integrations:** Live GoldAPI commodity rate tracker.
- **Deployment:** Continuous Integration on Vercel.

Slide 5: Core Database Architecture

- Real-time NoSQL schema design (Firestore).
- Optimized document structure for users, stores, login activity, audit logs, and catalog items.

Slide 6: Security & Access Control

- Role-Based Access Control (RBAC) across 7 roles.
- Automated account lockout after 5 failed login attempts.
- Cloud storage file validation rules.

Slide 7: Performance & UX Engineering

- Non-blocking login logging: Redirection speed reduced to **185ms**.
- Smart network timeouts (1.5s) to prevent page freezes.
- Responsive design system optimized for laptop screen widths (1366px–1440px).

Slide 8: Live System Demonstration

- Show customer selector flow, live rates catalog, and the admin branding dashboard.

Slide 9: Future Roadmap & Q&A

- Phase 2: Offline store POS sync, AI-driven jewellery virtual try-on, and automated supplier order flows.
- Open floor for questions.